Introduction to Data Analytics
MS - 4610

Session 6 – Inferential Statistics – 3

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Rejecting and failing to reject the null hypothesis

- Acceptance Matrix for hypothesis tests

		Decision	
		Reject The Null Hypothesis	Fail to Reject the Null hypothesis
Actual	Null hypothesis is true	Type 1 Error (or Producer Risk, False Positive, alpha-risk)	Correct Decision (1-alpha)
	Alternate Hypothesis is true	Correct Decision (Power = 1-Beta)	Type 2 Error (Consumer risk, False Negative, Beta risk)

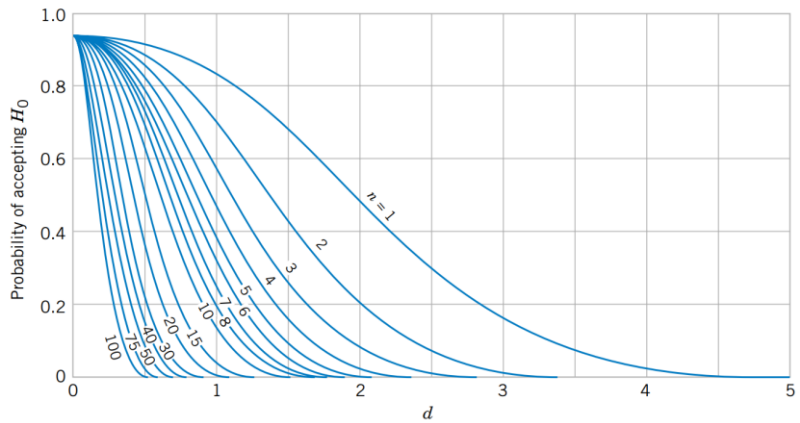
Type I and Type II Errors

- Prior to any data collection your type 1 error could be as high as alpha (α), and after analysis it is exactly equal to your *p*-value.

- Type II error (β) is more complicated. Why?
  - It is a function of Delta: $\delta = |\mu - \mu_0|$
  - It is a function of standard deviation: σ
  - Often we focus on $d = {}^{\delta}/_{\sigma}$
  - It is a function of Sample Size: *n*
  - It is a function of the Type I error: α

OC curves

- A graph of β versus d for a given sample size (n) is known as an OC (Operational Characteristic) curve:



(a) O.C. curves for different values of n for the two-sided normal test for a level of significance $\alpha = 0.05$.

Introduction

- So far statistical inference was confined to input variables that could take up two possible values (two sample tests), or there was no notion of an input variable (single sample tests).
- ANOVA
  - When there are three or more states of a single variable we can use ANOVA
- Chi-Square Test of Independence
  - Can be used when we want to compare multiple proportions

BASICS of ANOVA

- Tests the hypothesis that: $\mu_A = \mu_B = \mu_C = \mu_D$
- Why not multiple pairwise comparisons using t-tests?
- What do you do after a test? Tukey, Bonferroni, Scheffe tests
- Take the table:

	1	2	n
A	$y_{1,1}$	$y_{1,2}$	$y_{1,n}$
B	$y_{2,1}$	…	…
C	…	…	…
D	…	…	$y_{4,n}$

ANOVA OUTPUT

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F-Stat
Between Treatments	$n\sum_{i=1}^{a}(\bar{y}_{i.}-\bar{\bar{y}}_{..})^2$ or SSB	a-1	MSB = SSB/DoF	F = MSB/MSE
Error within treatments	$\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{i,j}-\bar{y}_{i.})^2$ or SSE	N-a	MSE = SSE/DoF	
Total	$\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{i,j}-\bar{\bar{y}}_{..})^2$ or SST	N-1	MST = SST/DoF	

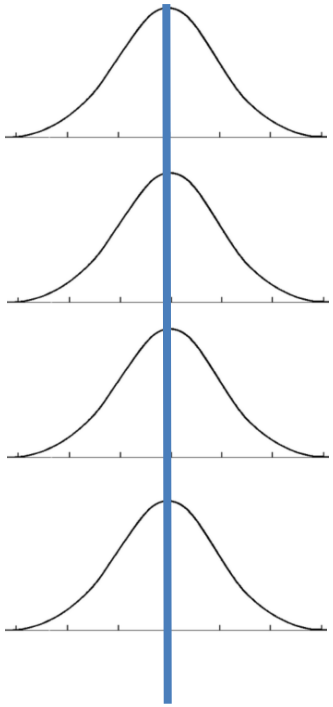
Compare F calculated against the F-distribution with a-1,N-a degrees of freedom and get a p-value

Why F for difference in means??

- The F is the ratio of two variances (where the samples come from a normal distribution and the null hypothesis is that the variances are equal)
- MSB is a way of calculating total variance
- MSE is a way of calculating total variance
- MSB, MSE and MST will be equal if the null hypothesis is true
- However is the null hypothesis is not, then MSB>MST>MSE

MSB and MSE

MSE



Total distribution of Y

Distribution of treatment A

Distribution of treatment B

Distribution of treatment C

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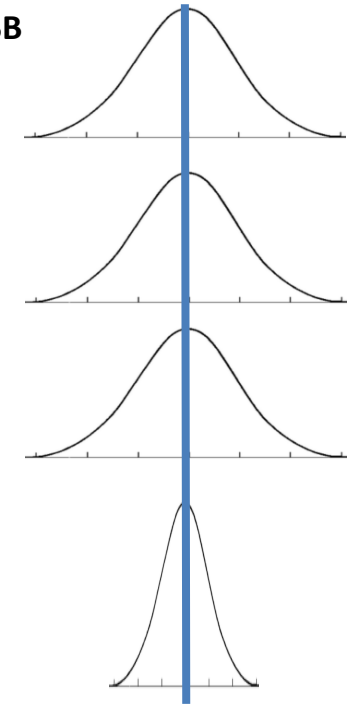
MSB



Total distribution of Y

Distribution of treatment A

Distribution of treatment B

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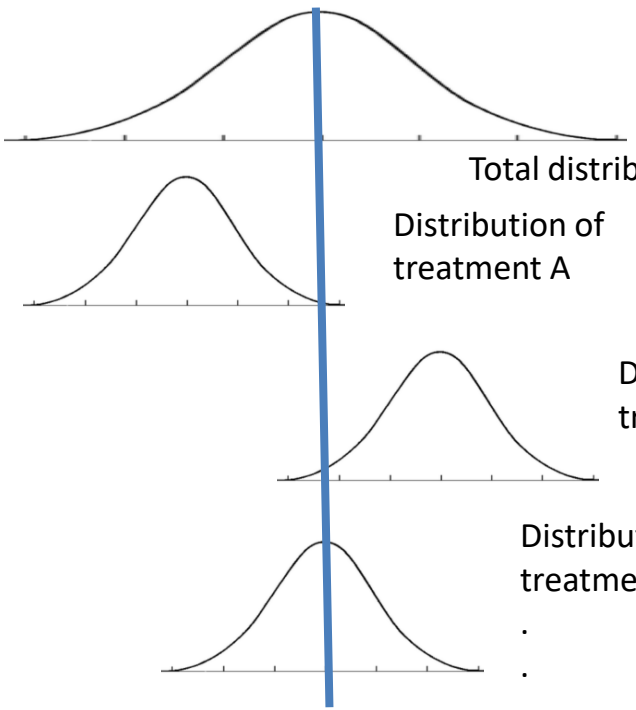
Distribution of the means of A,B,C

MSB and MSE

MSE



Total distribution of Y

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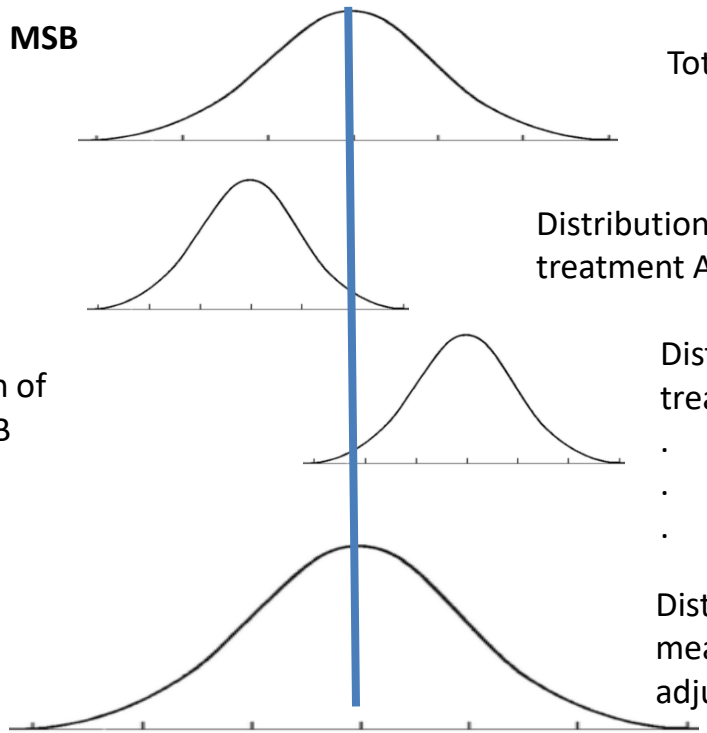
MSB



Total distribution of Y

Distribution of treatment A

Distribution of treatment B

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Distribution of the means of A,B,C adjusted by $\sqrt{n}$

Chi-Square TOI

- When using categorical variables
- Use this to test:
  - Does the input categorical variable effect the output categorical variable (works 2 or more states of the input or output variable)
  - Independence between two variables
  - Construct a contingency table:

Smoking habit / Exercise	Heavy	Regular	Occasional	Never
Frequent	7	9	12	87
Some	3	7	4	84
None	1	1	3	18

Chi Square TOI continued

- Create Theo values for this table in accordance to the assumption of independence
- It can be done row wise or column wise, but each cell gets an expected value
- Then if the null hypothesis is true then the test statistic is:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$$

With (r-1)*(c-1) degrees of freedom (or rc-c-r+1)